

Truth Has a Boundary: A Machine-Checked Coupling Between Truth and Confidence under Connectedness and Exact Separation

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Abstract

Probing studies read truth-related signals from language-model activations with continuous readouts. We ask what follows when a truth readout δ_F and a confidence signal c are treated as continuous fields on a connected representation space. The answer splits in two. Exact pointwise separation (confidence above a threshold τ wherever the answer is strictly true, and below τ wherever it is strictly false) forces the confidence level set $\{c = \tau\}$ into the truth boundary $\{\delta_F = 0\}$, with no topology at all. Connectedness, continuity and coverage make $\{c = \tau\}$ nonempty. Together they yield a query with confidence exactly τ and truth-distance exactly zero, so a threshold-inclusive truth guarantee is inconsistent with these premises, while a threshold-exclusive one is consistent but silent at that query. We derive quantitative, geometric, symmetry and discrete relaxations and machine-check the core results in Lean 4 with Mathlib, marking the few steps proved only on paper. Because the theorems need no empirical test, our experiments measure their demanding premise instead. In five open language models of up to 1.5B parameters, along interpolation paths from true city statements to their negations, two independently trained linear readouts are exactly separated on at most 13% of paths, and the smallest truth slack that restores separation has a median of 0.07 to 0.42 of the truth field’s scale; across these five models, probe accuracy and slack have the same rank ordering. The model’s own confidence is further from the premise (median slack 0.12 to 0.91). The coupled query itself is not observable with these paths, which confound truth with negation.

1 Introduction

Linear probes recover truth-related directions from language-model activations, and steering along such directions changes model behaviour (Marks & Tegmark, 2024; Azaria & Mitchell, 2023; Li et al., 2023; Zou et al., 2023; Bürger et al., 2024). These pipelines apply a continuous readout at arbitrary points of an activation space, including interpolated and edited activations that no prompt produced. Whether such readouts track semantic truth is contested (Farquhar et al., 2023; Levinstein & Herrmann, 2025); we do not settle that question. We ask a narrower one: *if* a truth readout and a confidence signal are continuous fields on a connected space, what must hold of them jointly?

The obstacle to a clean answer is that “a model cannot be both confident and right everywhere” is either false or vacuous unless every premise is stated. Existing inevitability results for hallucination are statistical or computability-theoretic (Kalai & Vempala, 2024; Xu et al., 2024; Karbasi et al., 2025) and say nothing about where in representation space the difficulty sits.

Our observation is that the question decomposes. *Alignment* of the two fields’ level sets is a pointwise consequence of separation: exact separation alone forces $\{c = \tau\} \subseteq \{\delta_F = 0\}$ (Proposition 1). *Existence* is where topology enters: connectedness, continuity of c and coverage make $\{c = \tau\}$ nonempty. The mathematics is

elementary, and we say so; its value is that each premise does one identifiable job, and that the conclusion mentions no safety guarantee and no threshold convention, so the different guarantee conventions become corollaries of one statement.

Concretely, combining the two halves gives the *boundary coupling theorem* (Theorem 3): some query has confidence exactly τ and lies exactly on the truth boundary. We then relax each premise and price the relaxation, and we mechanize the results in Lean 4.

Empirically, the theorem needs no test, since it holds whenever its premises hold. What can be measured is how far real readouts are from its demanding premise, exact separation, in the theory’s own unit, the truth slack of Theorem 5. In five open models the premise is met exactly on few or no paths, and the measured slack has the same rank ordering as probe accuracy across the five linear readouts (Table 2); the model’s own confidence is further from the premise (Table 3). Whether the coupled query can be observed directly is a separate question, and with the available paths the answer is no (Section 9.5).

Contributions.

1. **Boundary containment and coupling, with a guarantee taxonomy.** Exact separation forces $\{c = \tau\} \subseteq \{\delta_F = 0\}$; connectedness, continuous confidence and coverage make the left side nonempty, so some query has $c = \tau$ and $\delta_F = 0$. Hence a threshold-inclusive truth guarantee is inconsistent with these premises, and a threshold-exclusive guarantee is consistent but silent at that query (Section 4).
2. **Priced relaxations, mechanized.** Positive truth slack is forced under an inclusive guarantee, Lipschitz corridor and tube bounds hold around the coupled query, probe boundaries are hyperplanes exactly (linear) or to first order (differentiable), symmetry can replace coverage, and a discrete analysis shows that connectedness is what supplies the boundary query (Sections 5–7). Several of these transfer variants of the continuity argument of Bhatt et al. (2026) to two fields. All main statements are checked in Lean 4 (Table 1), and the few paper-only steps are marked.
3. **A measurement of the separation premise.** In five open models we estimate the empirical truth slack and the fraction of points violating separation, for trained linear and MLP readouts and for the model’s own confidence, with bootstrap intervals over paths. The results are descriptive (one data split, one fit per readout), and we show that the coupled query is not observable with true-to-negation paths (Section 9).

What we do not claim. The theorems do not bound error rates, do not assert that deployed models are continuous or that their reachable inputs form a connected set, and do not show that any probe tracks semantic truth. The coupled query may be an activation that no prompt produces (Section 10). The experiments do not test the theorems, which are mathematical consequences of their premises, and they do not show the coupled query in any model; they quantify how far five small models are from the separation premise on one family of statements, from a single data split and a single fit per readout. Only Theorem 14 concludes that strictly false answers occur, and it does so under its own probabilistic hypotheses.

2 Related work

Inevitability of hallucination. Kalai & Vempala (2024) prove that, for arbitrary facts not determinable from training data, a generatively calibrated language model’s hallucination rate is lower-bounded, with high probability, by roughly the fraction of facts seen exactly once in training (a Good–Turing missing-mass estimate) minus its miscalibration. Kalai et al. (2025) argue that hallucinations arise as classification errors and persist because evaluations reward guessing. Xu et al. (2024) prove inevitability for computable models by diagonalization, Banerjee et al. (2024) argue for it from undecidability, Karpowicz (2025) derive an impossibility for hallucination control from mechanism design, and Karbasi et al. (2025) show, in an identification-in-the-limit framework, that hallucination detection from positive examples alone is impossible for most language collections and becomes possible with labelled negative examples. These results are rate-level, distributional or computational. Ours is topological, conditions on a continuous confidence field over

a representation space, and exhibits at least one query at which the obstruction sits; for the deterministic results it yields exact uncertainty rather than confident falsehood.

Truth in representations. Truth-related signals are linearly decodable from activations (Marks & Tegmark, 2024; Azaria & Mitchell, 2023). Unsupervised probes have been proposed (Burns et al., 2023), but they can recover prominent non-truth features (Farquhar et al., 2023), and supervised probes can fail to generalize, including under negation (Levinstein & Herrmann, 2025). Bürger et al. (2024) report a two-dimensional truth subspace containing a polarity-sensitive direction whose sign flips under negation, which is the empirical motivation for our oddness premise (Section 6). Steering along learned directions changes truthfulness (Li et al., 2023; Zou et al., 2023), and concepts are often modelled as directions (Park et al., 2024), though some features are irreducibly multi-dimensional (Engels et al., 2025). We take one premise of this programme, that a probe defines a continuous field whose zero set separates its classes, and derive what it implies jointly with a confidence field. Confidence signals themselves are produced in several ways, including trained heads over activations (Kadavath et al., 2022) and entropy over sampled generations (Farquhar et al., 2024); our theorems apply only to confidence that is a continuous function of the representation.

Abstention and the inevitability of errors. Selective classification studies classifiers that may abstain, trading coverage against risk (Chow, 1970; El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017). The inclusive and exclusive guarantees of Definition 2 are pointwise analogues of its risk constraints, and abstaining on a region is exactly the escape from Theorem 3 that Section 10 describes, since removing an open region can disconnect the domain. Shafahi et al. (2019) show that, for data on certain manifolds, adversarial examples are unavoidable for any classifier, by concentration of measure; their result bounds the measure of points near a decision boundary, while ours exhibits a point on it under a separation premise.

Topological arguments in learning. Bhatt et al. (2026) use a related continuity argument on a connected prompt space to show that continuous, utility-preserving prompt-injection wrappers cannot be complete, and mechanize it in Lean 4, together with discrete, ε -relaxed, Lipschitz, boundary-dimension, multi-turn and stochastic variants. The present work transfers that argument and its variants to a pair of fields, truth and confidence; in particular Theorem 2, Theorem 7 and Proposition 13 restate lemmas of that development for the composite field δ_F . The mathematical object differs: that work constrains a single continuous transformation $D : \mathcal{X} \rightarrow \mathcal{X}$ (a defense wrapper) through one score function, whereas this paper studies the compatibility of two scalar fields $(\delta_F, c) : \mathcal{X} \rightarrow \mathbb{R}^2$ and the forced relationship between their level sets. The new elements are the level-set containment and two-field coupling, the threshold-convention taxonomy, the symmetry and hyperplane results, and the premise measurements. The strict-versus-weak threshold distinction also appears there, in the form that a fixed boundary point is not strictly below the threshold. Borsuk–Ulam-type arguments give impossibility results for replicable learning (Chase et al., 2024); our symmetry results are scalar relatives that need only the intermediate value theorem (Borsuk, 1933; Matoušek, 2003). Topological analyses of trained networks (Naitzat et al., 2020; Hensel et al., 2021) and geometric deep learning (Bronstein et al., 2021) study or engineer such structure; we derive constraints from it.

3 Setting

Let $\mathcal{X}$ be a topological space of queries as the model represents them (for example residual-stream states) and $\mathcal{A}$ a topological space of answers.

Definition 1 (Model, truth field). A *model* is a map $F = (a, c) : \mathcal{X} \rightarrow \mathcal{A} \times \mathbb{R}$ with answer $a(x)$ and scalar confidence $c(x)$. A *truth field* is a map $\delta : \mathcal{X} \times \mathcal{A} \rightarrow \mathbb{R}$, read as a signed truth-distance: negative on strictly true answers, positive on strictly false ones. Its zero set is the *truth boundary*, and the *realized truth field* is $\delta_F(x) = \delta(x, a(x))$.

Definition 2 (Premises). Fix a threshold $\tau \in \mathbb{R}$, a truth slack $\varepsilon \geq 0$ and a confidence margin $\gamma \geq 0$.

- (T) *Connectedness*: $\mathcal{X}$ is connected and nonempty.
- (C) *Continuity*: c is continuous (some results also require a and δ to be continuous; each statement says which).

(Cov $_{\varepsilon}$) *Coverage*: some x has $\delta_F(x) < -\varepsilon$ and some has $\delta_F(x) > \varepsilon$.

(Sep $_{\varepsilon,\gamma}$) *Separation at τ* : $\delta_F(x) < -\varepsilon \Rightarrow c(x) > \tau + \gamma$ and $\delta_F(x) > \varepsilon \Rightarrow c(x) < \tau - \gamma$ for all x . *Exact separation* is $\varepsilon = \gamma = 0$.

A *threshold-inclusive guarantee* is $c(x) \geq \tau \Rightarrow \delta_F(x) < 0$ for all x ; a *threshold-exclusive guarantee* is $c(x) > \tau \Rightarrow \delta_F(x) < 0$.

Status of the premises. Each premise is an idealization, and we flag the evidence against each. *(T)* holds for ambient spaces such as $\mathbb{R}^d$, which is where probes and steering vectors are applied. The set of activations reachable from actual prompts is finite, and contextual embeddings show clustered structure (Cai et al., 2021); Section 7 prices dropping (T) and says where a language model does and does not meet it. *(C)* holds for compositions of continuous layers but fails at argmax decoding. *Coverage* holds for any model that is sometimes right and sometimes wrong. *Exact separation* is pointwise and much stronger than statistical calibration (Guo et al., 2017); it idealizes a perfect truth probe used as confidence, and Section 9 measures how far trained readouts are from it. A linear probe $w^\top x - b$ is a continuous candidate for δ_F ; whether its zero set matches semantic truth away from training samples is an empirical premise, not a theorem.

Remark 1 (Biconditional separation). Some results use *biconditional* separation at $\frac{1}{2}$: $c(x) > \frac{1}{2} \Leftrightarrow \delta_F(x) < 0$ and $c(x) < \frac{1}{2} \Leftrightarrow \delta_F(x) > 0$. It implies exact one-way separation at $\frac{1}{2}$ (`strictCalibrated_to_epsCalibrated_zero_half`).

4 Boundary containment and coupling

The central result has two halves, one pointwise and one topological.

Proposition 1 (Boundary containment). *Let $\mathcal{X}$ and $\mathcal{A}$ be arbitrary sets, $F = (a, c)$ and δ arbitrary maps, and suppose exact separation (Sep $_{0,0}$) holds at τ . Then*

$$\{x \in \mathcal{X} : c(x) = \tau\} \subseteq \{x \in \mathcal{X} : \delta_F(x) = 0\}.$$

No topology, continuity or coverage is used. Connectedness, continuity of c and coverage are what make the left-hand set nonempty (Theorem 3).

Proof. If $c(x) = \tau$ and $\delta_F(x) < 0$, separation gives $c(x) > \tau$; if $\delta_F(x) > 0$, it gives $c(x) < \tau$. Both contradict $c(x) = \tau$, so $\delta_F(x) = 0$. This is the second half of the Lean proof of `boundary_coupling` (Appendix B); it is not stated there as a separate lemma. $\square$

Theorem 2 (Boundary existence). *Assume (T), that a and δ are continuous, and (Cov $_0$). Then some $x_0 \in \mathcal{X}$ has $\delta_F(x_0) = 0$.*

Proof. δ_F is continuous and takes both signs on a connected space, so the intermediate value theorem (Mathlib’s `IsPreconnected.intermediate_value2`) gives a zero. Lean: `model_truth_boundary_nonempty`. $\square$

Theorem 3 (Boundary coupling). *Assume (T), (C), (Cov $_0$) and exact separation (Sep $_{0,0}$) at τ . Then $\{c = \tau\}$ is nonempty, so some $x_0 \in \mathcal{X}$ satisfies*

$$c(x_0) = \tau \quad \text{and} \quad \delta_F(x_0) = 0.$$

Neither a nor δ needs to be continuous.

Proof. Separation turns the two coverage witnesses into points with $c > \tau$ and $c < \tau$; the intermediate value theorem on the connected space gives x_0 with $c(x_0) = \tau$, and Proposition 1 gives $\delta_F(x_0) = 0$. Lean: `boundary_coupling`; the proof term is in Appendix B. $\square$

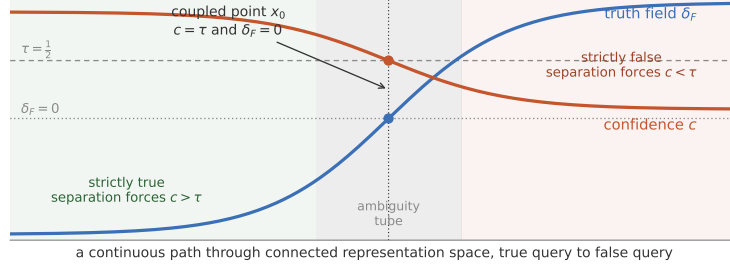


Figure 1: Schematic of Theorem 3 (illustrative curves, not data). Along a path from a strictly true to a strictly false query, exact separation forces confidence above τ wherever $\delta_F < 0$ and below τ wherever $\delta_F > 0$, so the two zero crossings coincide at the coupled query x_0 . The shaded band depicts the tube of Theorem 6; its width here is drawn by hand, not computed.

We call such an x_0 a *coupled query* (Figure 1). The theorem mentions no guarantee, so the choice between inclusive and exclusive guarantees only selects which corollary applies.

Corollary 4 (Guarantee taxonomy). *Under the hypotheses of Theorem 3:*

- (i) a threshold-inclusive guarantee at τ cannot also hold (Lean: `closed_guarantee_impossible`; the $\tau = \frac{1}{2}$, biconditional special case, stated with additional unused continuity hypotheses on a and δ , is `hallucination_trilemma`);
- (ii) the coupled query x_0 has $c(x_0) = \tau$, so it does not satisfy the antecedent $c(x_0) > \tau$ of a threshold-exclusive guarantee (Lean: `open_guarantee_silent`);
- (iii) a threshold-exclusive guarantee is consistent with all hypotheses (Example 1; proved on paper, not mechanized).

Example 1 (An exclusive guarantee is satisfiable). Let $\mathcal{X} = \mathcal{A} = \mathbb{R}$, $a(x) = x$, $\delta(x, y) = y$, $\tau = \frac{1}{2}$ and $c(x) = 1/(1 + e^x)$, a probability. Then $\mathcal{X}$ is connected, c is continuous, $\delta_F(-1) < 0 < \delta_F(1)$; since $c(x) > \frac{1}{2}$ exactly when $x < 0$, $\delta_F(x) < 0 \Leftrightarrow c(x) > \frac{1}{2}$ and $\delta_F(x) > 0 \Leftrightarrow c(x) < \frac{1}{2}$, so exact (indeed biconditional) separation holds. The exclusive guarantee holds because $c(x) > \frac{1}{2}$ implies $x < 0$, that is, $\delta_F(x) < 0$. The inclusive guarantee fails at $x = 0$, the coupled query. For a general threshold τ , the confidence $c(x) = \tau - \frac{1}{2} + 1/(1 + e^x)$ works the same way, since $c(x) > \tau$ exactly when $x < 0$.

The corollary is the paper’s trilemma. Coverage, exact separation and an inclusive guarantee cannot hold together on a connected space with continuous confidence. Replacing $\geq$ by $>$ in the guarantee removes the contradiction but not the coupled query.

5 Relaxations and the geometry of the boundary

Theorem 5 (Approximate coupling; slack must be positive). *Assume (T) , (C) , (Cov_ε) , $(Sep_{\varepsilon, \gamma})$ with $\varepsilon, \gamma \geq 0$, and a threshold-inclusive guarantee at τ . Then some x_0 has $c(x_0) = \tau$ and $-\varepsilon \leq \delta_F(x_0) < 0$. Consequently $\varepsilon > 0$: under these hypotheses exact truth slack is infeasible for every confidence margin $\gamma \geq 0$. Lean: `two_slack_approx_coupling`, `truth_slack_must_be_positive`; the $\varepsilon = 0$ case is `exact_from_approx`.*

Theorem 5 says only that $\varepsilon = 0$ is excluded. It does not show that any particular $\varepsilon > 0$ is achievable, and it gives no positive lower bound on the infimum of feasible slack.

Theorem 6 (Corridor and tube). *Let $\mathcal{X}$ be a pseudometric space, c L_c -Lipschitz and δ_F L_δ -Lipschitz. (i) If $L_c > 0$, any x, y with $c(x) \geq \tau + \gamma$ and $c(y) \leq \tau - \gamma$ satisfy $d(x, y) \geq 2\gamma/L_c$. (ii) If $c(x_0) = \tau$, $\delta_F(x_0) = 0$ and $L_c, L_\delta \geq 0$, then every x with $d(x, x_0) \leq r$ has $|\delta_F(x)| \leq L_\delta r$ and $|c(x) - \tau| \leq L_c r$. Lean: `confidence_gap_width`, `truth_margin_tube`, `ambiguity_tube`.*

Theorem 7 (Boundary sets). *For topological $\mathcal{X}, \mathcal{A}$ and continuous a, δ , the set $\{x : \delta_F(x) = 0\}$ is closed, and for every continuous $\gamma : \mathbb{R} \rightarrow \mathcal{X}$ and $s \leq t$ with $\delta_F(\gamma(s)) < 0 < \delta_F(\gamma(t))$ there is $u \in [s, t]$ with*

$\delta_F(\gamma(u)) = 0$. No connectedness of $\mathcal{X}$ is needed. Lean: `model_truth_boundary_isClosed`, `path_crosses_truth_boundary`.

Theorem 8 (Probe boundaries are hyperplanes, exactly or to first order). *Let E be a real vector space of finite dimension d . (i) Let $f : E \rightarrow \mathbb{R}$ be a nonzero linear map and $b \in \mathbb{R}$. Then $\{f = b\}$ is nonempty and equals $x_0 + \ker f$ for any x_0 with $f(x_0) = b$, and $\dim \ker f = d - 1$. (ii) Let E be normed, $f : E \rightarrow \mathbb{R}$ differentiable at x_0 with $f(x_0) = 0$ and derivative $L \neq 0$. Then every ball around x_0 contains points with $f > 0$ and points with $f < 0$; for every $\eta > 0$ there is $\rho > 0$ such that every x with $\|x - x_0\| < \rho$ and $f(x) = 0$ satisfies $|L(x - x_0)| \leq \eta \|x - x_0\|$; and $\dim \ker L = d - 1$.*

Proof. (i) Lean: `linear_probe_boundary_coset` (coset form, given a base point) and `linear_probe_boundary_dim` (rank-nullity). Nonemptiness is proved on paper: $f \neq 0$ gives some v with $f(v) \neq 0$, and $x_0 = (b/f(v))v$ satisfies $f(x_0) = b$. (ii) Lean: `regular_point_is_crossing`, `nonlinear_boundary_tangent` (which needs neither $L \neq 0$ nor finite dimension), `tangent_hyperplane_dim`. $\square$

For the probes used in practice, therefore, the zero set of δ_F that Theorem 3 places a coupled query on is an affine hyperplane, or near a regular point a set that deviates sublinearly from its tangent hyperplane.

6 Symmetry can replace coverage

Suppose negation acts on $\mathcal{X}$ as a continuous involution σ without fixed points and the realized truth field is odd, $\delta_F(\sigma x) = -\delta_F(x)$. This idealizes the polarity-sensitive truth direction reported by Bürger et al. (2024); probes trained on affirmative statements need not generalize to negations (Levinstein & Herrmann, 2025), and Section 9 measures the deviation from oddness.

Theorem 9 (Odd fields vanish). *If $\mathcal{X}$ satisfies (T) and carries such an involution σ , every continuous $g : \mathcal{X} \rightarrow \mathbb{R}$ with $g \circ \sigma = -g$ has a zero. The proof uses only oddness and the intermediate value theorem between x and σx . Lean: `antipodal_odd_has_zero`.*

Theorem 10 (Approximate and spherical oddness). *(i) If $\mathcal{X}$ satisfies (T), $\sigma : \mathcal{X} \rightarrow \mathcal{X}$ is any map, and g is continuous with $|g(\sigma x) + g(x)| \leq \varepsilon$ for all x , then some x has $|g(x)| \leq \varepsilon/2$. (ii) For $d \geq 2$, every function on $\mathbb{R}^d$ that is continuous on the unit sphere and odd there under $x \mapsto -x$ vanishes somewhere on the sphere. Lean: `approx_odd_near_zero`, `sphere_odd_has_zero`.*

Theorem 11 (Antipodal coupling). *Assume (T), a continuous fixed-point-free involution σ , continuous a , δ and c (the last is assumed in the Lean statement but unused), δ_F odd under σ , and biconditional separation at $\frac{1}{2}$. Then some x has $c(x) = \frac{1}{2}$ and $\delta_F(x) = 0$. No coverage hypothesis appears. Lean: `antipodal_hallucination_trilemma`.*

These are scalar relatives of Borsuk–Ulam; no Borsuk–Ulam theorem is used in the proofs. Theorem 10(ii) is stated for the Euclidean unit sphere only. Normalization layers map hidden states close to a sphere up to a learned affine map, but this connection is interpretation and is not formalized.

7 Without the continuum; turns and randomness

Token sequences are discrete. The following results show what remains when topology is removed: the boundary query must be assumed rather than derived.

Theorem 12 (Discrete impossibility). *For arbitrary sets $\mathcal{X}, \mathcal{A}$ and arbitrary F, δ with biconditional separation at $\frac{1}{2}$: if some query has $\delta_F = 0$, the inclusive guarantee at $\frac{1}{2}$ fails; and the inclusive guarantee holds if and only if no query has $\delta_F = 0$. Lean: `boundary_question_impossible`, `faithful_iff_boundary_free`.*

Proposition 13 (Discrete sign change). *If $g : \{0, \dots, n\} \rightarrow \mathbb{R}$ has $g(0) < 0 < g(n)$, some $i < n$ has $g(i) < 0 \leq g(i+1)$. Lean: `discrete_sign_change`.*

Along a sequence of token-level edits from a true to a false statement, δ_F therefore changes sign between adjacent steps, but no step need sit exactly on the boundary. Connectedness is what turns this straddling pair into a boundary query (Theorem 2).

Where a language model meets the premises. From its input embeddings onward, a transformer is a composition of continuous maps (linear layers, attention softmax, normalization, smooth activations), so as a function of its embeddings or hidden states it is continuous on a connected space. Probes, steering vectors, soft prompts and embedding-space attacks all act at this level, and there (T) and (C) hold without idealization. For models whose inputs are themselves continuous, such as images or audio, the actual input domain is a connected set (for example $[0, 1]^n$ for pixel intensities), so (T) holds for real inputs as well. Discreteness enters only at the text interface: the representations of actual prompts form a finite set for a bounded context length, so a coupled query need not be the representation of any prompt, and a guarantee stated only over prompts is not contradicted by Theorem 3. Theorem 6 bounds how much this helps. If a prompt’s representation lies within distance r of a coupled query, its confidence is within $L_c r$ of τ and its truth margin is at most $L_\delta r$, so prompts near a coupled query inherit near-ambiguity in proportion to their distance. How close real prompts come to coupled queries is an empirical question we do not measure here. In every case coverage and exact separation remain premises; settling the domain premise does not settle them.

Multiple turns. If a model maps a length- T history in $\mathcal{X}^T$ to one answer and confidence, $\mathcal{X}^T$ with the product topology is connected whenever $\mathcal{X}$ is. The $\frac{1}{2}$ -threshold biconditional form of Corollary 4(i) then applies, with the additional continuity of a and δ assumed in `multi_turn_history_dependent`. We have mechanized only this statement for histories; per-turn answer sequences are not modelled.

Randomness. If the *expected* confidence $\bar{c}$ and *expected* truth distance $\bar{\delta}$, viewed as functions of the query, satisfy (C), coverage and exact separation at $\frac{1}{2}$, Theorem 3 gives a query with $\bar{c} = \frac{1}{2}$ and $\bar{\delta} = 0$ (`stochastic_coupling_expected`). The Lean statement takes $\bar{c}, \bar{\delta}$ as arbitrary functions; deriving their continuity and separation from sample-level properties is not formalized. At such a query the following holds.

Theorem 14 (Stochastic dichotomy). *Let (Ω, μ) be a probability space and $c, d : \Omega \rightarrow \mathbb{R}$ with d integrable and $\int d d\mu = 0$. If $c \geq \frac{1}{2} \Rightarrow d < 0$ holds μ -almost everywhere and $\mu\{c \geq \frac{1}{2}\} > 0$, then $\mu\{d > 0\} > 0$. Lean: `stochastic_dichotomy`.*

Read with Ω the sampling randomness at a fixed query whose expected truth distance is zero, a sampler that is faithful on almost every sample and confident with positive probability must emit strictly false answers with positive probability. Instantiating Ω this way is a modelling step that we have not formalized. At such a query the hypothesis $\mu\{c \geq \frac{1}{2}\} > 0$ is automatic when c is integrable with mean $\frac{1}{2}$: otherwise $c < \frac{1}{2}$ almost surely, and its mean would be strictly below $\frac{1}{2}$ (paper proof, not mechanized).

8 Mechanization

The Lean 4 (de Moura & Ullrich, 2021) development, built on Mathlib (mathlib Community, 2020) (toolchain v4.28.0, Mathlib v4.28.0, 18 source files plus a root module) states results over arbitrary topological, pseudometric or normed spaces as each theorem requires. Table 1 maps every statement in this paper to its Lean identifier and marks whether it is mechanized (M) or proved on paper only (P).

A search of the source finds no `sorry`, `admit`, `native_decide` or `axiom` declarations. Several files contain `#print axioms` commands for main theorems, but we have not recorded their output for every identifier in Table 1, so we make no claim here about which axioms each theorem uses. The mechanized statements concern abstract spaces: the identification of $\mathcal{X}$ with a model’s representation space, of δ_F with a probe, and of σ with negation is modelling, not mechanized content.

9 Measuring the separation premise in language models

What is measured, and why. Theorem 3 holds whenever its premises hold, so it needs no empirical test. What can be measured is how far real readouts are from its one demanding premise, exact pointwise separation. The theory supplies the unit: Theorem 5 relaxes separation by a truth slack ε , and a readout pair is $(\varepsilon, 0)$ -separating exactly when every point with $|\delta_F| > \varepsilon$ has its confidence on the correct side of $\frac{1}{2}$. We therefore estimate, along interpolation paths in five language models, the smallest such ε , together with

Table 1: Statements and their machine-checked counterparts. M: mechanized in Lean; P: proved on paper only.

Result	Status	Lean identifier(s)
Thm. 2	M	<code>model_truth_boundary_nonempty</code>
Thm. 3	M	<code>boundary_coupling</code>
Cor. 4(i)	M	<code>closed_guarantee_impossible</code> ; <code>hallucination_trilemma</code> ($\tau = \frac{1}{2}$, biconditional, continuous a, δ)
Cor. 4(ii)	M	<code>open_guarantee_silent</code>
Cor. 4(iii)	P	Example 1
Thm. 5	M	<code>two_slack_approx_coupling</code> , <code>truth_slack_must_be_positive</code> , <code>exact_from_approx</code>
Thm. 6	M	<code>confidence_gap_width</code> , <code>truth_margin_tube</code> , <code>ambiguity_tube</code>
Thm. 7	M	<code>model_truth_boundary_isClosed</code> , <code>path_crosses_truth_boundary</code>
Thm. 8(i)	M / P	<code>linear_probe_boundary_coset</code> , <code>linear_probe_boundary_dim</code> ; nonemptiness on paper
Thm. 8(ii)	M	<code>regular_point_is_crossing</code> , <code>nonlinear_boundary_tangent</code> , <code>tangent_hyperplane_dim</code>
Thm. 9	M	<code>antipodal_odd_has_zero</code>
Thm. 10	M	<code>approx_odd_near_zero</code> , <code>sphere_odd_has_zero</code>
Thm. 11	M	<code>antipodal_hallucination_trilemma</code>
Thm. 12	M	<code>boundary_question_impossible</code> , <code>faithful_iff_boundary_free</code>
Prop. 13	M	<code>discrete_sign_change</code>
Multi-turn (Sec. 7)	M	<code>multi_turn_history_dependent</code> ($\frac{1}{2}$, biconditional, continuous a, δ)
Expected fields (Sec. 7)	M	<code>stochastic_coupling_expected</code> (fields assumed, not derived)
Thm. 14	M	<code>stochastic_dichotomy</code>
Prop. 1	M / P	inside the proof of <code>boundary_coupling</code> ; not a standalone lemma
Dichotomy hypothesis automatic (Sec. 7)	P	—
Rem. 1	M	<code>strictCalibrated_to_epsCalibrated_zero_half</code>

the fraction of points at which separation fails. These are measurements of a hypothesis, not tests of a prediction, so they need no null model. Section 9.5 asks the separate question of whether the coupled query itself is observable with this design; the answer is no.

9.1 Setup

Models: GPT-2 (124M), Pythia-410M, Qwen2.5-0.5B, Phi-1.5 (1.3B) and Qwen2.5-1.5B (Hugging Face ids in Appendix C). **Data:** the `cities` and `neg_cities` statements of Marks & Tegmark (2024), 1,496 statements each over 748 cities, each statement paired with its negation of opposite label, split by city into training split A (50%), split B (20%), validation (10%) and test (20%). **Representation:** the final-token hidden state at one layer, chosen by validation accuracy among $\{\lfloor n/4 \rfloor, \lfloor n/2 \rfloor, \lfloor 3n/4 \rfloor\}$ for an n -layer model. **Fields:** δ_F is the negated logit of a standardized logistic probe trained on split A. Confidence c is either a second logistic head trained on split B (*trained confidence*; the two readouts share no training statements but read the same representation), or the model’s own probability of the token “true” (*endogenous confidence*, Section 9.3). One training run of each readout was made. **Paths:** for each of the 149 test cities whose affirmative statement is true, the hidden state is interpolated linearly from that statement to its negation at 41 points. Paths whose endpoints the probe misclassifies are dropped, since the premise is only defined relative to a probe that gets the endpoints right. **Statistics, per path:** the *violation fraction*, the share of the 41 points where $\delta_F < 0$ but $c \leq \frac{1}{2}$, or $\delta_F > 0$ but $c \geq \frac{1}{2}$; whether the path is *exactly separated* (no violating point); and the *empirical truth slack* $\hat{\varepsilon}$, the largest $|\delta_F|$ among violating points divided by the endpoint scale $\max(|\delta_F(\text{start})|, |\delta_F(\text{end})|)$, which is the smallest ε for which $(\varepsilon, 0)$ -separation holds on the path ($\hat{\varepsilon} = 0$ if none). We report means and medians over paths with percentile bootstrap 95% intervals (10,000 path resamples), conditional on the trained readouts.

9.2 Trained confidence

RQ. How far from exact separation are two independently trained truth readouts of the same representation? **Result** (Table 2). Exact separation almost never holds on the grid: on 0% to 13% of paths. The premise fails

Table 2: Separation premise between two independently trained linear readouts. *Probe*, *Head*: test accuracy of δ_F and c . *Paths*: paths whose endpoints the probe classifies correctly, of 149. *Exact*: fraction of those paths with no violating point. *Violation*: mean fraction of violating points per path. $\hat{\varepsilon}$: median empirical truth slack, in units of the endpoint scale; 95% bootstrap intervals in brackets. Sources: `results.json`, `results_separation.json`.

Model	Layer	Probe	Head	Paths	Exact	Violation	$\hat{\varepsilon}$
Phi-1.5 (1.3B)	18/24	0.72	0.72	66/149	0.00	0.34 [0.29, 0.40]	0.42 [0.32, 0.57]
GPT-2 (124M)	9/12	0.77	0.72	83/149	0.02	0.29 [0.24, 0.34]	0.35 [0.21, 0.44]
Pythia-410M	18/24	0.82	0.77	104/149	0.00	0.27 [0.23, 0.32]	0.29 [0.21, 0.41]
Qwen2.5-0.5B	12/24	0.96	0.96	137/149	0.04	0.14 [0.12, 0.17]	0.16 [0.11, 0.18]
Qwen2.5-1.5B	14/28	1.00	1.00	148/149	0.13	0.07 [0.06, 0.08]	0.07 [0.05, 0.10]

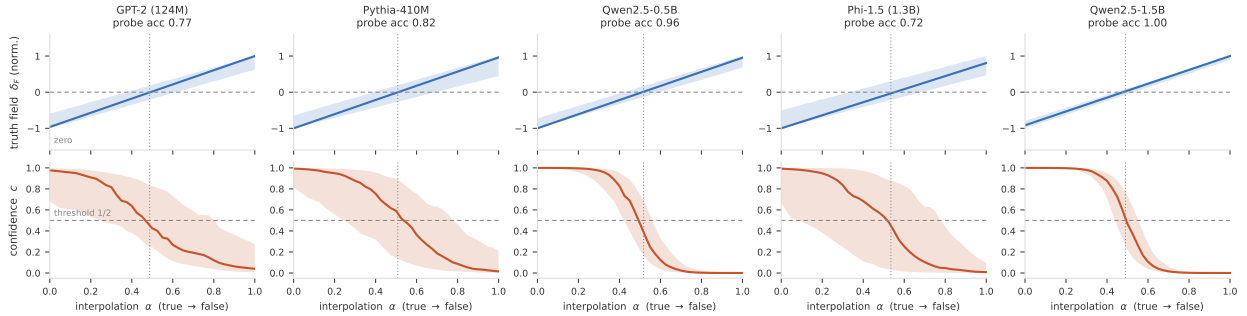


Figure 2: Trained linear readouts along true-to-negation paths (median and interquartile band over kept paths). Top: probe δ_F divided by the endpoint scale. Bottom: trained head c . Dotted line: mean position of the probe’s zero crossing. Under exact separation the bottom curve would cross $\frac{1}{2}$ exactly at the dotted line on every path; the spread around it is the separation violation measured in Table 2.

at 7% to 34% of path points on average, and the median empirical truth slack is 0.07 to 0.42 of the endpoint scale. Across the five models, probe accuracy, violation fraction and slack have the same rank ordering. This is a descriptive observation about five fits, not an estimated relationship, and the MLP readouts of Section 9.4 only partly preserve it. The slack distribution has a long tail: its 90th percentile is 0.20 for Qwen2.5-1.5B and at least 0.47 for the other four models, reaching the full endpoint scale for GPT-2 and Phi-1.5 (`results_separation.json`). **Relation to the theory.** The measured slack is strictly positive in every model, which is what Theorem 5 requires of any system that also carries an inclusive guarantee. That is consistency, not confirmation: these readouts carry no such guarantee, and positive slack would be expected of imperfect readouts in any case. **Does not establish.** Five models give a rank ordering, not a statistical trend, from a single data split and a single fit per readout. Parameter count is not the ordering variable: Phi-1.5 is the second-largest model but has the weakest probe, and a sweep over all its layers reaches at most 0.73 test accuracy.

Where separation fails. At the probe’s zero crossing, the trained confidence is usually far from $\frac{1}{2}$: only 17%, 10%, 11%, 5% and 21% of crossings (GPT-2, Pythia-410M, Qwen2.5-0.5B, Phi-1.5, Qwen2.5-1.5B) have $c \in [0.4, 0.6]$, and 31%, 43%, 40%, 67% and 17% have $c < 0.1$ or $c > 0.9$ (`results_crossing_conf.json`). The distribution is widely dispersed, and U-shaped for Pythia-410M, Qwen2.5-0.5B and Phi-1.5. The mean confidence at the crossing (0.52, 0.51, 0.49, 0.43, 0.54) is therefore not informative on its own.

9.3 The model’s own confidence

RQ. Both readouts in Section 9.2 are trained to predict truth. How far is the model’s own next-token confidence from the premise? **Setup.** Statements are wrapped in the template “Q: Is this statement true or false? ... A: The statement is”. We interpolate the last-token output of the final transformer block, patch it in, and read $p(\text{“true”})$ against $p(\text{“false”})$ through the final normalization and unembedding; the probe is

Table 3: Separation premise with endogenous confidence: the model’s own probability of the token “true” under a question template, read after patching interpolated final-block states. *Probe*: test accuracy of a probe trained on the same final-block states. *Paths*: paths whose endpoints that probe classifies correctly, of 149. *End. sep.*: paths whose confidence is on the correct side of $\frac{1}{2}$ at both endpoints. Other columns as in Table 2. Sources: `results_endogenous.json`, `results_separation.json`.

Model	Probe	Paths	End. sep.	Exact	Violation	$\hat{\epsilon}$
GPT-2 (124M)	0.66	53/149	23 (0.43)	0.00	0.41 [0.33, 0.49]	0.60 [0.28, 0.85]
Pythia-410M	0.76	84/149	0 (0.00)	0.00	0.49 [0.44, 0.54]	0.91 [0.79, 1.00]
Qwen2.5-0.5B	0.94	137/149	128 (0.93)	0.08	0.18 [0.15, 0.21]	0.18 [0.14, 0.25]
Phi-1.5 (1.3B)	0.78	80/149	53 (0.66)	0.00	0.41 [0.36, 0.46]	0.66 [0.53, 0.77]
Qwen2.5-1.5B	1.00	147/149	146 (0.99)	0.08	0.10 [0.09, 0.11]	0.12 [0.08, 0.16]

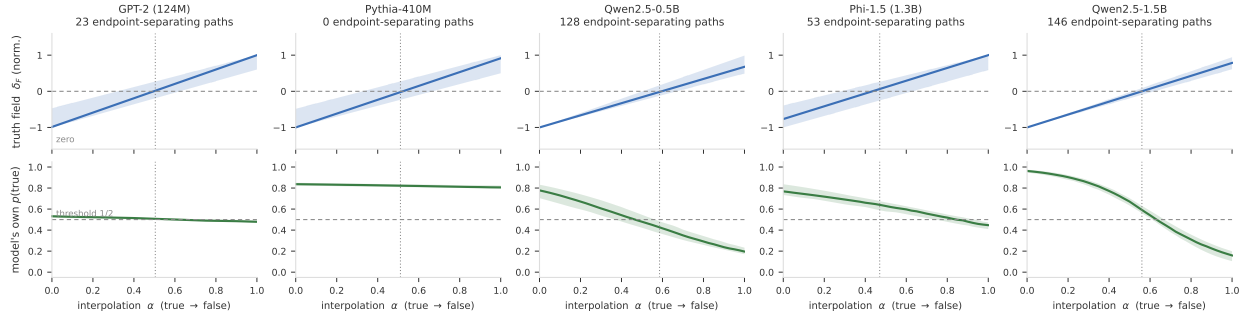


Figure 3: Endogenous confidence on endpoint-separating paths (all kept paths for Pythia-410M, which has none). Top: probe δ_F on final-block states, normalized by endpoint scale. Bottom: the model’s own $p(\text{true})$ on a common $[0, 1]$ axis. Dotted line: mean probe zero crossing. Only the two Qwen models sweep through $\frac{1}{2}$ near the probe’s crossing; Phi-1.5 crosses late, GPT-2 stays near $\frac{1}{2}$ throughout, and Pythia-410M stays above it.

trained on split A of the same final-block states. **Result** (Table 3, Figure 3). Endogenous confidence is further from the premise than trained confidence in every model. It comes closest in the two Qwen models (median slack 0.12 and 0.18, violation at 10% and 18% of points), which are also the only models whose confidence is on the correct side at both endpoints on more than 90% of paths. For Pythia-410M, whose confidence exceeds $\frac{1}{2}$ everywhere, the premise fails at about half of all points and the slack is close to the full endpoint scale. Phi-1.5 shows that endpoint agreement is a weak proxy for the premise: its confidence is on the correct side at both ends of 66% of paths, yet it crosses $\frac{1}{2}$ late on the path, and its slack (0.66) is the second largest. **Does not establish.** The endogenous confidence is a near-linear readout (normalization, then a two-token unembedding difference) of the same state the probe reads, so it is not independent of the representation.

9.4 Robustness

Nonlinear readouts. With one-hidden-layer MLP readouts (64 units, early stopping) on the same splits, probe accuracies are 0.79, 0.87, 0.97, 0.80 and 0.995 (GPT-2, Pythia-410M, Qwen2.5-0.5B, Phi-1.5, Qwen2.5-1.5B), and 98% or more of kept paths cross zero exactly once. The median slack is 0.35, 0.41, 0.21, 0.34 and 0.09, and the violation fraction 0.22, 0.29, 0.13, 0.26 and 0.06 (`results_separation.json`). The two Qwen models are again closest to the premise. Among the other three the order differs from the linear case: Pythia-410M has the most accurate MLP probe of the three but the largest slack.

Unmatched paths. For the linear readouts, δ_F and the logit of c are affine along a straight path, so every path is determined by the readout values at its endpoints; reconstructing the matched paths this way reproduces the saved curves exactly. This lets us build paths from the true affirmative of one test city to the negation of a different city, removing the matched pairing (4,290 to 21,756 paths per model;

`compute_crosspairs.py`). On these paths exact separation holds on 3% to 11%, and the median slack is 0.35, 0.26, 0.26, 0.16 and 0.09 for Phi-1.5, GPT-2, Pythia-410M, Qwen2.5-0.5B and Qwen2.5-1.5B (95% intervals from a bootstrap over cities: [0.27, 0.42], [0.20, 0.31], [0.22, 0.31], [0.14, 0.18], [0.07, 0.10]). Slack stays clearly positive in every model and is somewhat lower than on matched paths for the three weaker models. The rank agreement with probe accuracy holds except that GPT-2 and Pythia-410M become indistinguishable. These paths still flip truth and polarity together, because every stored statement is a true affirmative or a false negation, so this check addresses the matched pairing, not the confound of Section 9.5.

Oddness defect. Using each matched pair as $(x, \sigma x)$, the endpoint defect $|\delta_F(x) + \delta_F(\sigma x)| / \max(|\delta_F(x)|, |\delta_F(\sigma x)|)$ has median 0.40, 0.43, 0.30, 0.53 and 0.17, and 90th percentile 0.82, 0.91, 0.62, 0.89 and 0.41 (`results_oddness.json`). Theorem 10(i) needs a uniform bound on the defect, so these data do not support applying the symmetry results to these probes.

9.5 Can the coupled query be observed?

Under exact separation, the truth readout’s zero crossing and the confidence readout’s $\frac{1}{2}$ crossing would coincide on every path. With the measured slack they do not, and the distance between them, the *crossing gap*, has median 0.05 to 0.31 of the path for trained confidence (Appendix Table 4). We asked whether this gap reveals coupling beyond what the paths themselves impose, by comparing it, path by path, with a baseline readout that always crosses $\frac{1}{2}$ at the path midpoint and carries no truth information. On paths where the confidence is on the correct side at both endpoints, the trained or endogenous confidence is closer than the midpoint on fewer paths than it is farther, in all 14 model and readout conditions. In three conditions the difference is significant at a Bonferroni-corrected level over the 14 exact sign tests: MLP GPT-2 ($p = 0.0025$), MLP Qwen2.5-0.5B ($p = 0.0006$) and endogenous Phi-1.5 ($p = 0.0003$) (`results_paired.json`). It is never significant in the other direction. Two features of the design explain why this test is weak. First, for linear readouts on linear paths both fields are affine in the interpolation parameter, so both crossings are fixed by the endpoint values. Second, every path flips truth and negation together, so a readout of polarity alone would also cross on every path; in GPT-2 and both Qwen models a nearest-neighbour graph of the test states splits exactly by polarity, with true and false balanced inside each component (`results_gaps.json`). With the saved data, which contain only these paths and no hidden states, the coupled query is not observable. Paths from false affirmatives to true negations, same-polarity paths between true and false city attributions, and a polarity-probe readout would be needed.

Summary. In five models up to 1.5B parameters, the separation premise is met exactly on few or no paths. Its empirical slack is smallest, near 0.07 to 0.12 of the truth field’s scale, in the model with the most accurate probe, and across the five linear readouts slack and probe accuracy have the same rank ordering. The model’s own confidence is further from the premise than a trained head. The coupled query that the premise would force is not observable with these paths.

10 Limitations and how to falsify our interpretation

Scope of the theory. The theorems are existence statements about abstract spaces. They do not bound how often a model errs, and except for Theorem 14 they conclude uncertainty, not falsehood. They apply to a deployed system only through the modelling step that identifies its representation space, confidence and truth readout with $\mathcal{X}$, c and δ_F . For text inputs the coupled query may be an ambient activation no prompt produces; a guarantee quantified only over prompts is then not contradicted, Section 7 shows that a finite domain needs an assumed boundary query, and Theorem 6 only bounds the confidence and truth margin of prompts that lie near a coupled query. For continuous-input models this gap does not arise (Section 7). Abstaining on an open region removes it from the domain and can disconnect the true region from the false one, so refusal is an escape that the theorems explicitly allow. None of these escape routes is mechanized.

Scope of the experiments. The experiments cover five models of at most 1.5B parameters, one statement family (cities) with a translation replication for which only crossing gaps were saved, final-token states, three candidate layers, one data split, a single fit per readout, and linear true-to-negation interpolation paths that

confound truth with polarity. Separation statistics are computed on a 41-point grid and are conditional on paths whose endpoints the probe classifies correctly. The bootstrap intervals are conditional on the trained readouts. Hardware, software versions and runtimes were not recorded for the model runs; all statistics in Section 9 are recomputed from the released curves by scripts that need only NumPy.

What would weaken our interpretation. (i) The rank agreement between probe accuracy and slack would be undermined if it changed under other data splits, seeds or layers, or if it reversed on statement families other than cities. (ii) The observability question is open: on decoupled paths (false affirmative to true negation, or same-polarity attribution swaps), a confidence readout whose crossing tracks the truth readout’s crossing more closely than a midpoint or polarity baseline would be evidence that the coupled query is visible in practice, and its absence would be evidence that it is not. (iii) Evidence that probes used for steering are applied only on a provably disconnected set would remove the motivation for (T).

An open empirical question. The rank agreement in Table 2 suggests a hypothesis we have not tested: that the slack needed to make a truth readout and a confidence readout pointwise compatible falls as the truth readout becomes more accurate. Testing it requires many fits per model (seeds, splits and layers), several statement families, and nonlinear readouts, with probe accuracy and slack compared across fits rather than across five model averages. If it held, it would say that truth–confidence boundary compatibility improves as truth becomes more recoverable from the representation; if it failed, the ordering here would be an artifact of five particular fits.

Broader impact statement

The results concern the internal consistency of truth and confidence readouts. They could be misread as showing that language models “must lie”; outside Theorem 14 they do not, and we have tried to state each premise where it is used. A possible use is diagnostic: a coupled query may indicate where a probe-based confidence signal is least informative; we have not evaluated this use.

11 Conclusion

Exact separation forces the confidence threshold set into the truth boundary; connectedness, continuous confidence and coverage make that set nonempty, so the two must meet at a query. That one fact settles both guarantee conventions: an inclusive guarantee is inconsistent, and an exclusive one is silent at the meeting point. We mechanized the result and its relaxations in Lean 4 and marked what remains on paper. In five small models the separation premise is met exactly on few or no paths; across the five linear readouts its empirical slack has the same rank ordering as probe accuracy, and it is larger for the model’s own confidence. Observing the coupled query directly needs paths that decouple truth from negation, which is the next experiment.

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A Lean conventions

Results are stated for $Q \vdash A : \text{Type}^*$ with topologies, a model $M : Q \rightarrow A \times \mathbb{R}$ and a truth field $\delta : Q \times A \rightarrow \mathbb{R}$. The realized field is $\text{fun } q \Rightarrow \delta(q, (M\ q).1)$ and confidence is $(M\ q).2$. The conditions in Definition 2 correspond to `EpsCovering M delta eps` (coverage), `TwoSlackSeparating M delta c eps gamma` (separation), whose equal-slack case `EpsCalibrated M delta c eps` is definitionally equal (`epsCalibrated_iff_twoSlack`), `StrictCalibrated M delta` (biconditional separation at $\frac{1}{2}$) and `TrilemmaFaithful M delta` (inclusive guarantee at $\frac{1}{2}$). The inclusive guarantee at a general threshold appears inline as a hypothesis. The exclusive guarantee has no Lean definition. Lipschitz conditions are explicit inequalities over a `PseudoMetricSpace`, and the hyperplane results use `LinearMap` with rank-nullity.

B Core proof term

The Lean proof of Theorem 3, with ASCII notation (`intermediate_value2` renders Mathlib’s `intermediate_value2`).

```
theorem boundary_coupling
  {Q A : Type*} [TopologicalSpace Q] [ConnectedSpace Q]
  [TopologicalSpace A]
```

```

(M : Q -> A x R) (delta : Q x A -> R) (c : R)
(hconf : Continuous (fun q => (M q).2))
(hcov : EpsCovering M delta 0)
(hcal : EpsCalibrated M delta c 0) :
exists q0, (M q0).2 = c /\ delta (q0, (M q0).1) = 0 := by
obtain <<qt, ht>, <qf, hf>> := hcov
have hct : (M qt).2 > c + 0 := hcal.1 qt ht
have hcf : (M qf).2 < c - 0 := hcal.2 qf hf
have hct' : (M qt).2 > c := by linarith
have hcf' : (M qf).2 < c := by linarith
obtain <q0, _, hq0> :=
  isPreconnected_univ.intermediate_value2
    (mem_univ qf) (mem_univ qt)
    hconf.continuousOn continuous_const.continuousOn
    (le_of_lt hcf') (le_of_lt hct')
refine <q0, hq0, ?_>
by_contra hne
rcases lt_or_gt_of_ne hne with hlt | hgt
. have hd : delta (q0, (M q0).1) < -0 := by simp using hlt
  have := hcal.1 q0 hd
  linarith
. have := hcal.2 q0 hgt
  linarith

```

C Experimental details

Models. Hugging Face ids gpt2, EleutherAI/pythia-410m, Qwen/Qwen2.5-0.5B, microsoft/phi-1_5 and Qwen/Qwen2.5-1.5B, run in float32.

Splits. The 748 cities are shuffled with a fixed seed and split into test (20%), validation (10%), split B (20%) and split A (50%); each city contributes its affirmative statement and its negation. Scalars are fit on training data only, and test cities are held out of both readouts.

Layer selection. Validation accuracy among three candidate layers. Layers index Hugging Face `hidden_states`, where index 0 is the embedding output and index n the final block. For Phi-1.5, a separate sweep over layers 1–24 selects layer 23 on validation (0.75), with best test accuracy 0.727 at layer 24; the main run uses layer 18 from the three-candidate rule.

Readouts. Logistic regression (lbfgs) on standardized features. The probe and head differ only in their training split.

Relative boundary width. The median of $|\delta_F|$ over path points with $c \in (0.4, 0.6)$, divided by the endpoint scale: 0.33, 0.36, 0.20, 0.37 and 0.11. This is a descriptive proxy, not the formal slack ε of Definition 2, which is a supremum-type quantity.

Crossing gaps. Table 4 reports the crossing gap used in Section 9.5: the distance in interpolation parameter between $\arg \min |\delta_F|$ and $\arg \min |c - \frac{1}{2}|$ on the grid, for the trained linear readouts. Under exact separation it would be zero. The permuted-label and random-direction nulls (20 seeds each) rarely cross $\frac{1}{2}$ inside a path, so their gaps sit near an endpoint by construction and they are weak baselines; the midpoint baseline of Section 9.5 is the informative comparison.

Translation statements. On the 354 Spanish–English translation statements of Marks & Tegmark (2024), split by statement rather than by word (so the same word can appear in training and test statements), with layers fixed from the main run, probe accuracy is 0.94 and 0.98 for Qwen2.5-0.5B and Qwen2.5-1.5B, 0.70

Table 4: Crossing gaps for trained linear readouts. *Gap*: median over kept paths, with bootstrap 95% interval. *Gap|sep*, *Mid|sep*: median gap and midpoint baseline on paths where the head is on the correct side at both endpoints. *Perm/Rand*: median null gaps. Sources: `results_bootstrap.json`, `results_controls.json`, `results_endogenous.json`.

Model	Gap [95% CI]	Gap sep	Mid sep	Perm / Rand
GPT-2 (124M)	0.250 [0.175, 0.325]	0.175	0.125	0.475 / 0.325
Pythia-410M	0.200 [0.150, 0.287]	0.150	0.100	0.450 / 0.450
Qwen2.5-0.5B	0.100 [0.100, 0.125]	0.100	0.100	0.425 / 0.400
Phi-1.5 (1.3B)	0.312 [0.175, 0.375]	0.175	0.200	0.481 / 0.381
Qwen2.5-1.5B	0.050 [0.050, 0.075]	0.050	0.050	0.450 / 0.450

and 0.71 for Pythia-410M and Phi-1.5, and 0.52 (chance) for GPT-2. Only crossing gaps were saved for this run, not the path curves, so the separation statistics above cannot be computed for it. The gaps (0.100 and 0.050 for the Qwen models on 38 and 41 paths; 0.525 and 0.325 for Pythia-410M and Phi-1.5 on 18 and 19 paths) follow the same order as on cities. Because the separation statistics of Section 9 cannot be computed for this run, it does not replicate the paper’s main measurement, and we report it only here.

Exploratory adversarial run. We froze an MLP truth readout and trained a 256–256 MLP confidence head on all non-test statements plus fresh interpolants. The hinge losses targeted the separation premise and an inclusive guarantee, with weight 5 on the guarantee. Training was 60 full-batch Adam steps (learning rate 10^{-3}); NumPy was seeded but PyTorch was not, so head initialization is not reproducible. The fraction of evaluation points violating the guarantee was 0.005, 0.016, 0.001, 0.018 and 0.002 at the end, so the inclusive-guarantee hypothesis of Theorem 5 does not hold for any trained head, and the run does not test that theorem. The empirical slack estimate (the largest $-\delta_F$ among true points with $c \leq \frac{1}{2}$, relative to endpoint scale) ended at 0.91, 0.80, 0.38, 0.57 and 0.22 and had not converged for several models (`results_adversarial.json`). We report this run for completeness only.

D Reproducibility

The supplementary material contains the Lean project (toolchain and Mathlib pinned at v4.28.0; build with `lake build`), the experiment scripts, the public datasets of Marks & Tegmark (2024), all result files, and the plotting scripts that regenerate Figures 2 and 3 from them. `compute_separation.py`, `compute_crosspairs.py`, `compute_crossing_conf.py`, `compute_controls.py` and `compute_paired.py` compute every statistic of Section 9 from the saved curves, without model runs. Package versions used for the model runs were not recorded.